

Stage 3: 3D Pose Lifting (Design Specification)

Overview

Stage 3 converts temporally ordered 2D human poses into temporally consistent 3D pose sequences using a monocular temporal model (e.g., MotionAGFormer). This stage focuses strictly on signal reconstruction and does not perform interpretation, metrics, or football logic.

Pipeline Position

Stage 1: ActionInstance → Detection & Tracking → PlayerTrackSets

Stage 2: PlayerTrackSets → 2D Pose Estimation → PoseTrackSets

Stage 3: PoseTrackSets → 3D Pose Lifting → Pose3DTrackSets

Stage 4: Pose3DTrackSets → Metrics / Stability / Football Logic

Responsibilities

- Accept PoseTrackSet objects per camera view
- Process each PoseTrack independently
- Lift 2D joints to 3D using temporal context
- Preserve frame alignment and track identity
- Output numeric 3D joint tensors only

Non-Goals

- No multi-view fusion
- No player selection or prioritization
- No visualization or video output
- No world or field coordinate estimation
- No football-specific logic

Input Contract

Input is a PoseTrackSet per camera view. Each PoseTrack contains an ordered list of Pose2D objects. Each Pose2D includes a frame_id and 17 COCO-format keypoints (x, y, confidence). Frames may be missing and confidence values may vary.

Output Contract

Output is a Pose3DTrackSet per camera view. Each Pose3DTrack includes track_id, status (VALID / INSUFFICIENT_LENGTH / FAILED), and an ordered list of Pose3D objects. Each Pose3D includes frame_id and 17 joints (x, y, z). Coordinates are relative and camera-centric.

Core Assumptions

Monocular 3D pose lifting does not recover true depth. It estimates the most plausible 3D configuration given 2D motion and learned human motion priors. Temporal context is mandatory.

Root Joint Definition

The root joint is defined as the midpoint between left_hip and right_hip keypoints in COCO format. All poses are normalized relative to this root.

Temporal Processing

Default temporal window size is fixed at 27 frames. Non-overlapping windows are used by default. Windows with insufficient valid frames are skipped.

Coordinate Normalization

Before inference, poses are centered on the root joint. Optional scale normalization may be applied depending on model configuration. Camera orientation is preserved.

Confidence Handling

Default behavior uses only x and y coordinates for inference. Keypoints with confidence below a threshold (e.g., 0.2) are masked. Continuous confidence weighting is not applied at this stage.

Model Inference

A temporal transformer model such as MotionAGFormer is used. Input shape is $(T \times 17 \times 2)$ or $(T \times 17 \times 3)$, output shape is $(T \times 17 \times 3)$. The model is treated as a pure inference function.

Edge Cases & Safeguards

Short tracks are marked as INSUFFICIENT_LENGTH. Inference failures are marked as FAILED and do not crash the pipeline. No interpolation is performed at this stage.

Confidentiality

No raw frames, images, or videos are saved. Only numeric 3D pose tensors are produced and persisted.

Completion Criteria

Stage 3 is complete when PoseTrackSet objects are reliably converted to Pose3DTrackSet objects with clean contracts, no architecture violations, and readiness for downstream analysis.

Downstream Compatibility

Outputs are designed for stability metrics, motion analysis, and football analytics in later stages. Stage 3 itself remains strictly isolated.